

AYUSH BHATTACHARYA

AI Engineering | Agentic Workflows | Banking Automation

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github.com/Ayushbh6

PROFESSIONAL SUMMARY

AI developer with experience building document review platforms, RAG applications and automation for banking risk teams. Built ALIGNIA from the ground up with Python, FastAPI and Next.js and led its development team through live pilot testing with a pharmaceutical company. At Erste Group, own the Databricks workspace and build AI tools and data workflows for operational-risk reporting. MSc in Quantitative Finance.

PROFESSIONAL EXPERIENCE

Operational Risk Intern

Erste Group Bank AG, Vienna, Austria | Jan 2026 - Present

- Manage the team's Databricks workspace and SQL workflows for operational-loss reporting and investigations.
- Built a one-click large-loss review pipeline that screens the daily loss-event flow (tens to hundreds of events) across a three-level entity hierarchy, replacing a manual daily review and returning checked results in about a minute.
- Deployed a FastAPI review application and task-specific AI agents as Databricks apps used directly by team members.
- Built an AI summariser that drafts loss-event descriptions for the loss database and a classifier now being moved into production on the loss-data platform.
- Run ad-hoc stress tests on the Advanced Measurement Approach (AMA) model, quantifying the operational-risk capital and risk-weighted asset (RWA) impact of entity scenarios.



Lead AI Developer

AI by DNA, Remote | May 2025 - Dec 2025

- Built ALIGNIA from the ground up to review pharmaceutical promotional materials against Greek and European compliance frameworks, company procedures and scientific references.
- Led a team of three interns for around six months, then delivered the platform with one developer, reporting directly to the CEO.
- Platform entered live pilot testing with a European pharmaceutical company during the engagement.
- Developed workflows for claim extraction, evidence retrieval and reference verification, with human review, version tracking and exportable reports.



AI Developer

Integration Technologies Group, Inc., Fairfax, Virginia | Jan 2025 - May 2025

- Delivered a text-to-SQL agent with built-in verification in an agentic harness; ITG client testers reported same-model intent-to-SQL accuracy rising from roughly 60–70% to over 95%.
- Developed multi-agent and retrieval-augmented generation (RAG) applications for financial and compliance work.
- Integrated enterprise documentation into AI applications and built Python/FastAPI services.



Analyst, Financial Advisory

Deloitte Austria, Vienna, Austria | Oct 2023 - Dec 2024

- Developed and validated credit-risk models in R and SAS, including Loss Given Default (LGD) and Expected Credit Loss (ECL) calculations.
- Automated financial-data processing with Python and supported model validation, governance and work involving Basel and IFRS requirements.



Summer Intern, Financial Advisory

Deloitte Austria, Vienna, Austria | Jul 2023 - Sep 2023

- Developed Probability of Default (PD) and LGD models in R and migrated legacy SAS credit-risk models to R.

Senior Analyst, Business Consulting

Ernst & Young LLP, Bengaluru, India | Oct 2020 - Jun 2022

- Supported asset-management implementation for a pharmaceutical company and conducted Sarbanes-Oxley (SOX) and Internal Financial Controls (IFC) audits.
- Performed on-site inventory reconciliation for a beverage company.

Deloitte.



SELECTED AI PROJECTS

Socrates - AI Coding and Investigation Workspace

Built a local AI workspace with model integrations, tool execution, persistent history and context compression for extended tasks.

github.com/Ayushbh6/Socrates

DPA Guru - AI Contract Risk Review

Built AI-assisted contract review with document ingestion, hybrid retrieval and page-level evidence for reviewing data-processing agreements.

github.com/Ayushbh6/DPA_Guru

SEC Summariser - Financial Filing Analysis

Built an AI research application that retrieves SEC filings, preserves financial tables and supports document-based analysis.

github.com/Ayushbh6/SEC-Summariser

TECHNICAL SKILLS

AI and retrieval

LLM applications, multi-agent workflows, RAG, document processing, evidence retrieval, OpenAI and Gemini APIs.

Backend and applications

Python, FastAPI, TypeScript, JavaScript, Next.js, REST APIs, Docker, Git and GitHub Copilot in VS Code.

Data

SQL, Databricks, PostgreSQL, pgvector, MongoDB, SQLite, pandas and NumPy.

EDUCATION

Master of Science in Quantitative Finance

Vienna University of Economics and Business (WU) | Sep 2022 - Nov 2024

Bachelor of Science in Economics, Mathematics and Statistics

Christ University, Bengaluru | Jun 2017 - Jun 2020



CERTIFICATIONS

Advanced Credit Risk Management - TU Delft

Managerial Economics and Business Analysis - University of Illinois at Urbana-Champaign

LANGUAGES

English (Professional) | German (A2, actively learning) | Hindi and Bengali (Native)